Ticket #006B: Missing Default Gateway

Symptoms: DHCP successfully assigns addressing parameters but does not install the default gateway into the routing table.

Resolve:

1. Investigating dhclient.leases to determine exactly what the DHCP server offered the machine.

```
jay@mainserver:~$ cat /var/lib/dhcp/dhclient.leases
cat: /var/lib/dhcp/dhclient.leases: No such file or directory
jay@mainserver:~$
```

Analysis: This server uses a different network manager than dhclient.

2. Checking whether the network manager received a default gateway but did not apply it.

```
jay@mainserver:~$ nmcli dev show enp0s1
Command 'nmcli' not found, but can be installed with:
```

Analysis: The server is not using NetworkManager.

3. Identifying what is managing networking.

```
jay@mainserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 32:18:e8:f7:66:da brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.228/24 metric 100 brd 192.168.1.255 scope global dynamic enp0s1
       valid_lft 26211sec preferred_lft 26211sec
    inet6 2603:6080:6300:13a4::1314/128 scope global dynamic noprefixroute
```

Analysis: DHCP successfully assigns addressing information, but the default gateway option is not installed in the routing table by the client.

4. Narrowing down to what service is managing DHCP by checking the logs of systemd-networkd.

```
jay@mainserver:~$ systemctl status systemd-networkd
● systemd-networkd.service - Network Configuration
     Loaded: loaded (/usr/lib/systemd/system/systemd-networkd.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-04-15 16:41:59 EDT; 18h ago
TriggeredBy: ● systemd-networkd.socket
       Docs: man:systemd-networkd.service(8)
             man:org.freedesktop.network1(5)
   Main PID: 475 (systemd-network)
     Status: "Processing requests..."
      Tasks: 1 (limit: 4312)
   FD Store: 0 (limit: 512)
     Memory: 3.5M (peak: 3.9M)
        CPU: 143ms
     CGroup: /system.slice/systemd-networkd.service
             └─475 /usr/lib/systemd/systemd-networkd

Apr 15 16:41:59 mainserver systemd-networkd[475]: lo: Link UP
Apr 15 16:41:59 mainserver systemd-networkd[475]: lo: Gained carrier
Apr 15 16:41:59 mainserver systemd-networkd[475]: Enumeration completed
Apr 15 16:41:59 mainserver systemd[1]: Started systemd-networkd.service - Network Configuration.
Apr 15 16:41:59 mainserver systemd-networkd[475]: enp0s1: Configuring with /run/systemd/network/10-netplan-enp0s1.network.
Apr 15 16:41:59 mainserver systemd-networkd[475]: enp0s1: Link UP
Apr 15 16:41:59 mainserver systemd-networkd[475]: enp0s1: Gained carrier
Apr 15 16:42:01 mainserver systemd-networkd[475]: enp0s1: Gained IPv6LL
Apr 15 16:42:02 mainserver systemd-networkd[475]: enp0s1: DHCPv6 address 2603:6080:6300:13a4::1314/128 (valid for 6d 23h 59min 59s, prefer
59min 59s)
Apr 15 16:42:03 mainserver systemd-networkd[475]: enp0s1: DHCPv4 address 192.168.1.228/24, gateway 192.168.1.1 acquired from 192.168.1.1
```

Analysis: It is clear now that DHCP provides the gateway and systemd-networkd receives it.

5. Now that systemd-networkd is identified as the manager, the next step is to determine whether the failure to implement the default gateway in the ip route table is due to kernel-level routing behavior or a systemd-networkd configuration issue.

```
jay@mainserver:~$ networkctl status enp0s1
● 2: enp0s1
                     Link File: /usr/lib/systemd/network/99-default.link
                  Network File: /run/systemd/network/10-netplan-enp0s1.network
                         State: routable (configured)
                  Online state: online
                          Type: ether
                          Path: pci-0000:00:01.0
                        Driver: virtio_net
                        Vendor: Red Hat, Inc.
                         Model: Virtio network device
              Hardware Address: 32:18:e8:f7:66:da
                           MTU: 1500 (min: 68, max: 65535)
                         QDisc: fq_codel
  IPv6 Address Generation Mode: eui64
      Number of Queues (Tx/Rx): 1/1
              Auto negotiation: no
                       Address: 192.168.1.228 (DHCP4 via 192.168.1.1)
                                2603:6080:6300:13a4::1314
                                2603:6080:6300:13a4:3018:e8ff:fef7:66da
                                fd00:209a:7de7:1eaa:3018:e8ff:fef7:66da
                                fe80::3018:e8ff:fef7:66da
                       Gateway: 192.168.1.1
                                fe80::229a:7dff:fee7:1eaa
                           DNS: 192.168.1.1
                                2603:6080:6300:13a4::1
                Search Domains: lan
             Activation Policy: up
           Required For Online: yes
               DHCP4 Client ID: IAID:0xf1f5dd7f/DUID
             DHCP6 Client IAID: 0xf1f5dd7f
             DHCP6 Client DUID: DUID-EN/Vendor:0000ab11d68c58e3d458a047

Apr 15 16:41:59 mainserver systemd-networkd[475]: enp0s1: Configuring with /run/systemd/network/10-netplan-enp0s1.network.
Apr 15 16:41:59 mainserver systemd-networkd[475]: enp0s1: Link UP
Apr 15 16:41:59 mainserver systemd-networkd[475]: enp0s1: Gained carrier
lines 1-35
```

Analysis: All DHCP configuration, including the default gateway, is successfully applied by systemd-networkd. The interface is reported as routable and online, indicating correct network manager configuration and alignment with kernel-level routing. The issue is fully resolved.

Conclusion: No persistent misconfiguration was identified. The issue appears to have been a transient routing state condition that was fixed after manual intervention and network stabilization.